

ANCESTRAL CHARACTER ESTIMATION

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Phylogeography Workshop

Day 1

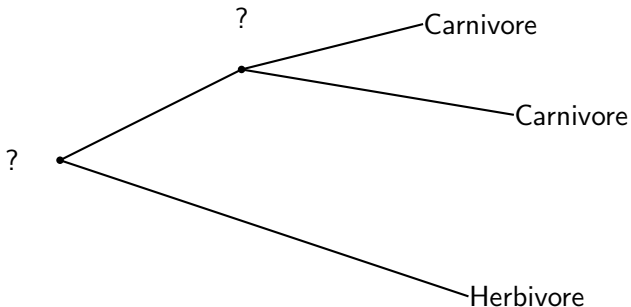
- ① BACKGROUND FOR ANCESTRAL CHARACTER ESTIMATION
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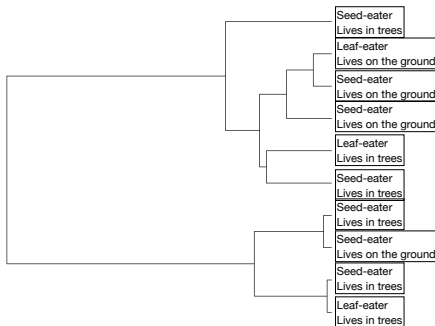
WHY DO ANCESTRAL CHARACTER ESTIMATION?

In evolutionary biology, we might ask: how did a current set of traits evolve?



THE HISTORICAL DEVELOPMENT OF THESE METHODS:

Originally, the methods we are about to discuss were developed for understanding **correlated evolution of ≥ 2 traits**¹



Example: Is evolution of diet correlated with habitat?

¹Mark Pagel. "Detecting correlated evolution on phylogenies: a general method for the comparative analysis of discrete characters". In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 255.1342 (1994), pp. 37–45.

A VARIETY OF METHODS FOR ANCESTRAL RECONSTRUCTION

A number of different methods to infer ancestral states exist:

- parsimony
- Markov models of discrete state changes
- Brownian motion or Ornstein-Uhlenbeck for continuous traits changing over time (but we won't discuss in this workshop)

PARSIMONY

Parsimony is intuitive and seems reasonable at first glance

There are some undesirable outcomes from this method

- multiple changes might occur over long periods of time
- parsimony reconstructions may not be unique, and it is also difficult to quantify uncertainty in ancestral inferences

A DIFFERENT APPROACH: USING MARKOV MODELS

If traits evolve in a neutral fashion, then we could model them in the same way that we imagine nucleotides in genomes change over time

For discrete traits, we postulate a rate of change from each trait value to another

MARKOV CHAINS UNDERLIE PHYLOGENETIC RECONSTRUCTION

Nucleotide changes over time are modeled with continuous-time Markov chains .. why not do this with discrete traits as well?

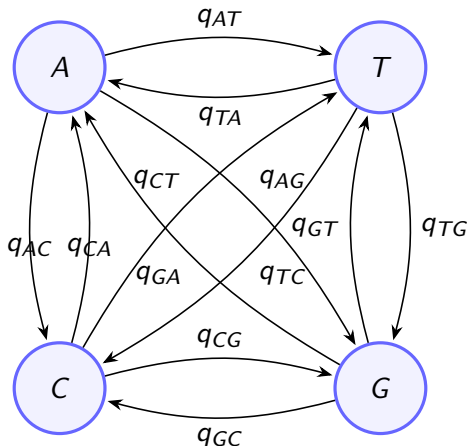


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ESSENTIAL FEATURES OF MARKOV CHAINS

If you worked through *Simulating nucleotide substitution models* on the website, you may have familiarized yourself with some of the mathematics of Markov chains

Briefly, we will discuss some of the essential features together

TRANSITION PROBABILITIES

Suppose we have a simple 2-state Markov chain with transitions occurring in continuous time

We let $P_i(t)$ represent the probability that the chain resides in state i at time t

In our simple 2-state case, we can write ordinary differential equations (ODEs) describing rates of change in state probabilities:

$$\begin{aligned}\frac{dP_1}{dt} &= -q_{12}P_1 + q_{21}P_2, \\ \frac{dP_2}{dt} &= q_{12}P_1 - q_{21}P_2.\end{aligned}$$

TRANSITION PROBABILITIES

$$\begin{aligned}\frac{dP_1}{dt} &= -q_{12}P_1 + q_{21}P_2, \\ \frac{dP_2}{dt} &= q_{12}P_1 - q_{21}P_2.\end{aligned}$$

In the R demo for nucleotide substitution models, we show how to solve this system by hand. There are two key steps

- recognize that $P_1 + P_2 = 1$ to eliminate a variable
- use an integrating factor to solve the one-dimensional ODE that is left

TRANSITION PROBABILITIES

In the demo, we see that solutions to the system

$$\begin{aligned}\frac{dP_1}{dt} &= -q_{12}P_1 + q_{21}P_2, \\ \frac{dP_2}{dt} &= q_{12}P_1 - q_{21}P_2,\end{aligned}$$

have the form

$$\begin{aligned}P_1(t) &= P_{1,0}e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right), \\ P_2(t) &= P_{2,0}e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right).\end{aligned}$$

TRANSITION PROBABILITIES

$$P_1(t) = P_{1,0}e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right),$$
$$P_2(t) = P_{2,0}e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right).$$

Examining the behavior of this solutions, we verify that
 $P_i(0) = P_{i,0}$ (satisfies initial conditions)

We also immediately see the long-term behavior:
 $\lim_{t \rightarrow \infty} P_i(t) = \frac{q_{ji}}{q_{ij}+q_{ji}}$

TRANSITION PROBABILITIES

$$P_1(t) = P_{1,0}e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right),$$
$$P_2(t) = P_{2,0}e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right).$$

Then, we can calculate transition probabilities $p_{11}(t)$, $p_{12}(t)$, $p_{21}(t)$, and $p_{22}(t)$ by supplying the appropriate initial conditions:

$$p_{11}(t) = P_1(t) \text{ with } P_{1,0} = 1, \quad p_{12}(t) = P_2(t) \text{ with } P_{2,0} = 0,$$
$$p_{21}(t) = P_1(t) \text{ with } P_{1,0} = 0, \quad p_{22}(t) = P_2(t) \text{ with } P_{2,0} = 1$$

TRANSITION PROBABILITIES

So, (check that) the transition probabilities $p_{ij}(t)$ are

$$\begin{pmatrix} \frac{q_{12}}{q_{12}+q_{21}} e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12}+q_{21}} & \frac{q_{12}}{q_{12}+q_{21}} (1 - e^{-(q_{12}+q_{21})t}) \\ \frac{q_{21}}{q_{12}+q_{21}} (1 - e^{-(q_{12}+q_{21})t}) & \frac{q_{21}}{q_{12}+q_{21}} e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12}+q_{21}} \end{pmatrix}$$

Useful checks: $p_{11}(0) = 1$, $p_{22}(0) = 1$, $p_{12}(0) = 0$, and $p_{21}(0) = 0$ because state changes cannot occur at time $t = 0$

STOCHASTIC RATE MATRIX

If we go back to the original ODEs,

$$\begin{aligned}\frac{dP_1}{dt} &= -q_{12}P_1 + q_{21}P_2, \\ \frac{dP_2}{dt} &= q_{12}P_1 - q_{21}P_2,\end{aligned}$$

we can re-write this system as

$$\frac{dP}{dt} = QP,$$

where

$$Q = \begin{pmatrix} -q_{12} & q_{21} \\ q_{12} & -q_{21} \end{pmatrix}$$

and $P = (P_1, P_2)^T$.

STOCHASTIC RATE MATRIX

Writing Q like this:

$$Q = \begin{pmatrix} -q_{12} & q_{21} \\ q_{12} & -q_{21} \end{pmatrix},$$

is using the convention that q_{ij} is the transition rate from state i to state j .

STOCHASTIC RATE MATRIX

Sometimes, the entries of Q are written like this,

$$Q = \begin{pmatrix} -q_{21} & q_{12} \\ q_{21} & -q_{12} \end{pmatrix},$$

so that q_{ij} is the transition rate into state i from state j .

Pay careful attention to this when working in R!

STOCHASTIC RATE MATRIX

$$Q = \begin{pmatrix} -q_{12} & q_{21} \\ q_{12} & -q_{21} \end{pmatrix},$$

A stochastic rate matrix like this always has the property that rates in columns sum to zero

Thus, all stochastic rate matrices have a nullspace, i.e., a zero eigenvalue, making them singular

Stochastic rate matrices also have negative terms on the diagonal. The sum of the diagonal equals sum of the eigenvalues – so we know that the sum of the eigenvalues is negative

STOCHASTIC RATE MATRIX

$$Q = \begin{pmatrix} -q_{12} & q_{21} \\ q_{12} & -q_1 \end{pmatrix},$$

In this example, the nullspace of Q , $\mathcal{N}(Q)$, is spanned by the vector

$$P^* = \begin{pmatrix} q_{21} \\ q_{12} \end{pmatrix}.$$

If we normalize this vector, we get

$$P^* = \begin{pmatrix} \frac{q_{21}}{q_{12}+q_{21}} \\ \frac{q_{12}}{q_{12}+q_{21}} \end{pmatrix}.$$

STOCHASTIC RATE MATRIX

$$P^* = \begin{pmatrix} \frac{q_{21}}{q_{12} + q_{21}} \\ \frac{q_{12}}{q_{12} + q_{21}} \end{pmatrix}.$$

This vector is the same one that describes the long-term behavior of the system

$$P_1 = P_{1,0}e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right),$$
$$P_2 = P_{2,0}e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12} + q_{21}} \left(1 - e^{-(q_{12}+q_{21})t}\right).$$

STOCHASTIC RATE MATRIX

This is no coincidence. The normalized vector P^* satisfying

$$\frac{dP^*}{dt} = QP^* = 0$$

corresponds to the **stationary distribution** of the Markov chain

Regardless of the initial conditions, in the long term limit, the probability of finding the Markov chain in a particular state is described by the stationary distribution

MASTER EQUATION

In general, can always obtain the probabilities of a Markov chain by solving the equation

$$\frac{dP}{dt} = QP.$$

CALCULATING TRANSITION PROBABILITIES

The general solution is

$$P(t) = e^{Qt} P_0,$$

where P_0 is the vector of initial state probabilities at time 0, and e^{Qt} is the matrix exponential operation defined by

$$e^{Qt} = I + Qt + \frac{1}{2!}(Qt)^2 + \frac{1}{3!}(Qt)^3 + \dots \quad (1)$$

In R, the function `expm` (from the package of the same name) can calculate this numerically. So, in practice, we just need the stochastic rate matrix, Q , and an initial condition P_0

TRANSITION PROBABILITIES

The transition probabilities $p_{ij}(t)$ are in fact entries of the **Fundamental Solution Matrix**, $\Phi(t) = e^{Qt}$:

$$\begin{pmatrix} \frac{q_{12}}{q_{12}+q_{21}} e^{-(q_{12}+q_{21})t} + \frac{q_{21}}{q_{12}+q_{21}} & \frac{q_{12}}{q_{12}+q_{21}} (1 - e^{-(q_{12}+q_{21})t}) \\ \frac{q_{21}}{q_{12}+q_{21}} (1 - e^{-(q_{12}+q_{21})t}) & \frac{q_{21}}{q_{12}+q_{21}} e^{-(q_{12}+q_{21})t} + \frac{q_{12}}{q_{12}+q_{21}} \end{pmatrix}$$

(Check that $\Phi(0) = I$)

DOES A STATIONARY DISTRIBUTION ALWAYS EXIST?

There is always a 0 eigenvalue of the matrix Q

This means rows/columns are not independent, so we can always find a vector in the nullspace, $\mathcal{N}(Q)$

If $\mathcal{N}(Q)$ is spanned by a single vector, then the stationary distribution is unique because of the requirement that $\sum_i P_i = 1$ (assuming some other requirements on Q)

PROPERTIES GUARANTEEING A UNIQUE STATIONARY DISTRIBUTION

Irreducibility: for a continuous-time Markov chain, there is a positive probability of transitioning from one state to any other (the state space is not composed of disjoint sets having transitions only within themselves)

Positive recurrence: the expected time for the Markov chain to return to any particular state is finite (guaranteed by irreducibility in a finite state space, but not in an infinite state space)

MARKOV MODELS FOR TRAIT EVOLUTION

Continuous-time Markov chains (CTMC) on finite state spaces are fundamental to statistical phylogenetics, phylogeography, and phylodynamics

$$Q = \begin{pmatrix} -\lambda_{12} & \lambda_{21} \\ \lambda_{12} & -\lambda_{21} \end{pmatrix}$$

Can we use these models to understand evolutionary histories?

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MARKOV MODELS FOR DISCRETE STATES ON TREES

Markov chains are useful for reconstructing phylogenies from genomic data, but let's start with a simpler problem first (and actually, one which arises frequently in phylogeographic analysis/ancestral state reconstruction):

Suppose we already have a phylogenetic tree and information about discrete traits at the tips (the data we collect). How do we quantify the probabilities of different scenarios of trait evolution, starting at the most recent common ancestor, down to our sampled/observed individuals?

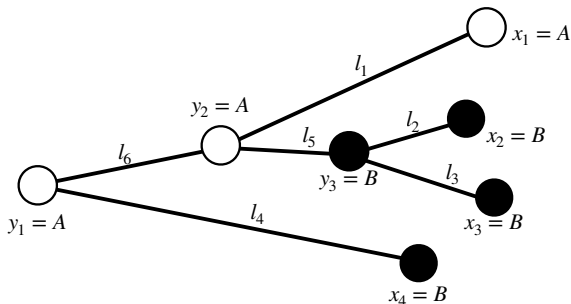
THE APPROACH TO RECONSTRUCTING ANCESTRAL STATES

Ancestral state reconstruction relies on basic ideas from statistics.

- 1 Propose a probabilistic model for trait evolution on the tree
 - 1A The model needs to integrate/sum over over all possible (unobserved) evolutionary scenarios down to the present day
- 2 Fit the model to the data by estimating parameters governing rates of trait change over time
 - 2A Use Maximum Likelihood to choose the best parameter estimates
- 3 Identify the (unobserved) evolutionary scenario that maximizes the probability of the data we observe under the model

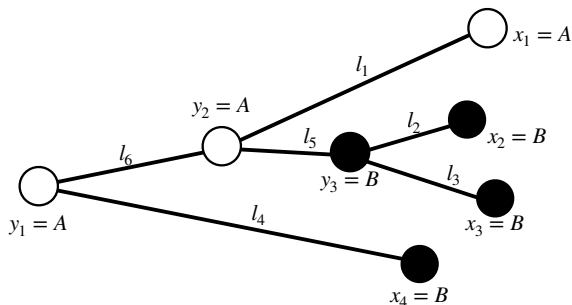
MARKOV MODELS FOR DISCRETE STATES ON TREES

Suppose we are given a phylogenetic tree, $\mathcal{T}$, we have information about tip locations, x_i , and we superimpose a continuous-time Markov chain with two states (A and B) on the tree



If we knew the locations of internal nodes, y_i , we could calculate transition probabilities along each branch and combine these to calculate the probability of observing the tip data

MARKOV MODELS FOR DISCRETE STATES ON TREES



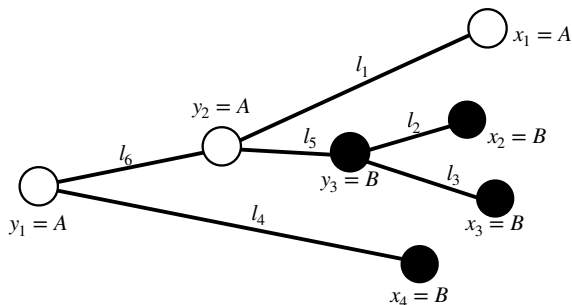
Example: transition from $y_1 = A$ to $x_4 = B$:

$$p(x_4|y_1) =$$

$$p(X_4 = B|Y_1 = A, l_4, Q) = e^{Ql_4} P_0,$$

with $P_0 = (1, 0)^T$

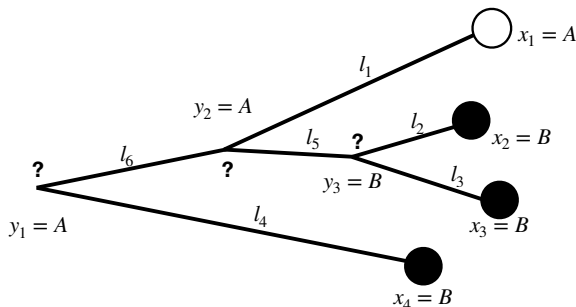
MARKOV MODELS FOR DISCRETE STATES ON TREES



Repeating this along each branch, we could calculate the probability of observing the joint configuration of tip states, x_i , and node states, y_i :

$$P(\vec{x}, \vec{y} | \mathcal{T}) = \pi(y_1) p(x_4 | y_1) p(y_2 | y_1) p(x_1 | y_2) p(y_3 | y_2) p(x_2 | y_3) p(x_3 | y_3)$$

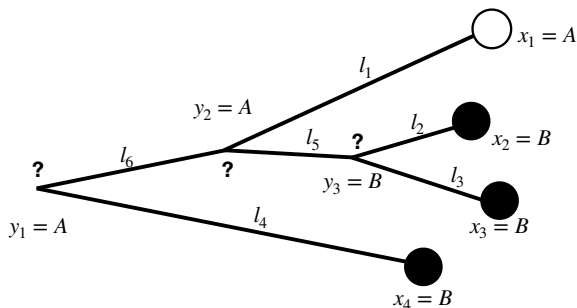
CALCULATING LIKELIHOOD



However, there is a problem: we don't actually observe/know the the values of internal nodes, y_i , so we have to sum over all possible configurations of the internal nodes:

$$P(\vec{x}|\mathcal{T}) = \sum_{\vec{y}} \pi(y_1)p(x_4|y_1)p(y_2|y_1)p(x_1|y_2)p(y_3|y_2)p(x_2|y_3)p(x_4|y_3)$$

CAN FIT TO DATA USING MAXIMUM LIKELIHOOD



The transition probabilities $p(y_j|y_i)$, $p(x_k|y_j)$ depend on parameters, θ , of a continuous-time Markov chain that we specify; if we interpret $P(\vec{x}|\mathcal{T}, \theta)$ as a function of θ , then we can fit the Markov model to the data $(\mathcal{T}, \vec{x})$ using Maximum Likelihood

THE TREE STRUCTURE ALLOWS US TO “PRUNE”

How do we calculate $p(\mathbf{x}|\mathcal{T})$? The likelihood can be calculated using the *pruning* (or, dynamic programming) method:

$$\sum_{\vec{y}} \pi(y_1) p(x_4|y_1) p(y_2|y_1) p(x_1|y_2) p(y_3|y_2) p(x_2|y_3) p(x_4|y_3) =$$

$$\sum_{y_1} \pi(y_1) p(x_4|y_1) \times$$

$$\sum_{y_2} p(y_2|y_1) p(x_1|y_2) \times$$

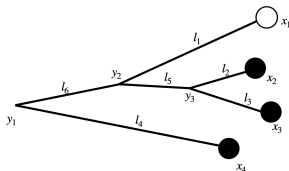
$$\sum_{y_3} p(y_3|y_2) p(x_2|y_3) p(x_3|y_3)$$

The calculation is carried from right to left, starting with the marginalization in y_3 , then in y_2 , then in y_1 . This makes the calculation feasible for large trees

PRUNING

The form of the pruned likelihood indicates that we calculate from the tips toward the root. First, we calculate the likelihood of the tree descending from node y_3 , conditional on the state being z ,

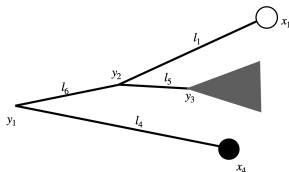
$$L_{y_3}(z) = p(x_2|y_3 = z, l_2)p(x_3|y_3 = z, l_3) :$$



PRUNING

The notation $L_{y_k}(z)$ refers to the likelihood of the tree descending from node y_k , given that y_k has state z .

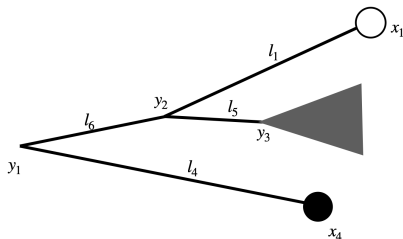
So we have as many terms $L_{y_3}(z)$ as there are possible values, z , for the node y_3



PRUNING

Once we have $L_{y_3}(z)$, we can calculate $L_{y_2}(z)$:

$$L_{y_2}(z) = p(x_1|y_2 = z, l_1) \left(\sum_w p(y_3 = w|y_2 = z, l_5) L_{y_3}(w) \right)$$

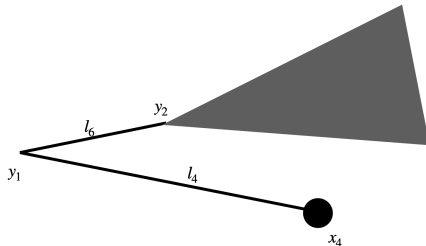


PRUNING

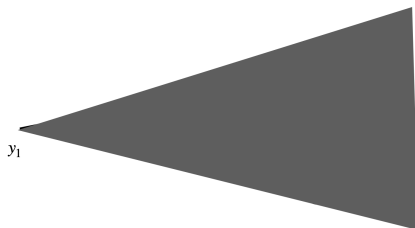
Finally, we get to $L_{y_1}(z)$:

$$L_{y_1}(z) = p(x_4|y_1 = z, l_4) \left(\sum_u p(y_2 = u|y_1 = z, l_6) L_{y_2}(u) \right),$$

and we have contributions for each of the possible values of $y_1 = z$.



PRUNING



To obtain the tree likelihood, we combine the likelihoods of the tree conditional on y_1 being in state z . This is accomplished by setting

$$L_{y_1} = \sum_{y_1} \pi(z) L_{y_1}(z).$$

for some distribution π describing the probability that the root is in state z .

PRUNING

$$L_{y_1} = \sum_z \pi(z) L_{y_1}(z).$$

In ace, the distribution π is chosen to be the stationary distribution of the Markov chain (which ensures a detailed balance condition holds throughout the tree whenever the Markov model is reversible)

The $\pi(y_1)$ is necessary for the tree likelihood to be interpreted as a marginalization of a joint distribution; there is no Bayesian interpretation here (or at least, there does not need to be)

PRUNING

$$L_{y_1} = \sum_{y_1} \pi(y_1) L_{y_1}(z).$$

In the phytools functions (e.g. `make.simmap`), the distribution π can be the stationary distribution of the Markov chain like `ace`, but can also be the “FitzJohn root prior”², or a user-supplied distribution

For reversible Markov chains (irreducible, aperiodic, and positive-recurrent), selecting π to be the stationary distribution ensures detailed balance throughout the phylogeny

²Richard G FitzJohn, Wayne P Maddison, and Sarah P Otto. “Estimating trait-dependent speciation and extinction rates from incompletely resolved phylogenies”. In: *Systematic biology* 58.6 (2009), pp. 595–611.

MARKOV MODELS FOR BINARY TRAIT EVOLUTION

Fun side note: in the original presentation of these models, Pagel was interested to study correlated evolution of two or more traits.

If the traits evolve in a correlated manner, state changes between the four combinations of states can be modeled with a 4×4 Markov generator (up to 12 parameters):

$$Q = \begin{pmatrix} -\sum_{j \neq 1} \lambda_{1j} & \lambda_{12} & \lambda_{13} & \lambda_{14} \\ \lambda_{21} & -\sum_{j \neq 2} \lambda_{2j} & \lambda_{23} & \lambda_{24} \\ \lambda_{31} & \lambda_{32} & -\sum_{j \neq 3} \lambda_{3j} & \lambda_{34} \\ \lambda_{41} & \lambda_{42} & \lambda_{43} & -\sum_{j \neq 4} \lambda_{4j} \end{pmatrix}$$

MARKOV MODELS FOR BINARY TRAIT EVOLUTION

If, on the other hand, the traits evolve independently of each other (“null hypothesis”), state changes between the four combinations of states are modeled with two independent Markov chains, each with a 2×2 Markov generator (4 parameters):

$$Q^{(1)} = \begin{pmatrix} -\lambda_{12}^{(1)} & \lambda_{21}^{(1)} \\ \lambda_{12}^{(1)} & -\lambda_{21}^{(1)} \end{pmatrix} \quad Q^{(2)} = \begin{pmatrix} -\lambda_{12}^{(2)} & \lambda_{21}^{(2)} \\ \lambda_{12}^{(2)} & -\lambda_{21}^{(2)} \end{pmatrix}$$

If the more complicated model fits significantly better (using, e.g., a likelihood ratio test), that suggests trait evolution is indeed correlated.

MARKOV MODELS FOR BINARY TRAIT EVOLUTION

For our purposes, we will primarily focus on evolution of individual traits. The overall goal is to understand how we can use the model to estimate the most likely configuration of unobserved ancestral states, given the data and the model.

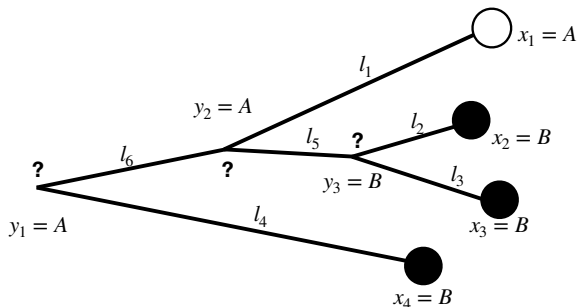
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ANCESTRAL STATE RECONSTRUCTION

To reconstruct ancestral states, we use the likelihood function we just obtained. Let's examine the form of the likelihood a little more carefully:

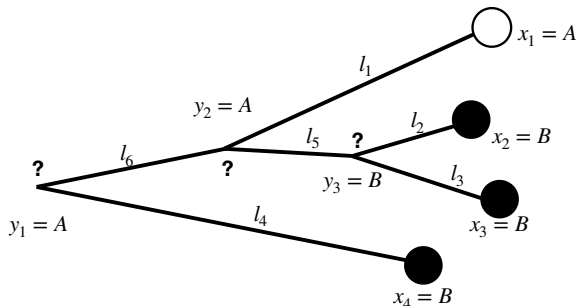
MARKOV MODELS FOR DISCRETE STATES ON TREES



The **likelihood** of the model is the probability of observing the tip states, $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ given the branch lengths l_i and parameters of the Markov model in $\mathbf{Q}$:

$$P(\vec{x}|\mathcal{T}) = \sum_{\mathbf{y}} \pi(y_1) p(x_4|y_1) p(y_2|y_1) p(x_1|y_2) p(y_3|y_2) p(x_2|y_3) p(x_4|y_3)$$

CALCULATING LIKELIHOOD



Because don't know the the states of internal nodes, y_i , we have to sum over all possible configurations, $\mathbf{y}$ (and we saw that we could use the “pruning algorithm” to do this efficiently):

$$P(\vec{x}|\mathcal{T}) = \sum_{\mathbf{y}} \pi(y_1) p(x_4|y_1) p(y_2|y_1) p(x_1|y_2) p(y_3|y_2) p(x_2|y_3) p(x_4|y_3)$$

THE LIKELIHOOD IS A MARGINALIZATION OVER LATENT VARIABLES

Even though there are a lot of conditionals ($p(y_k|y_j)$) in the expression of the likelihood,

$$P(\vec{x}|\mathcal{T}) = \sum_{\mathbf{y}} \pi(y_1)p(x_4|y_1)p(y_2|y_1)p(x_1|y_2)p(y_3|y_2)p(x_2|y_3)p(x_4|y_3),$$

don't be confused – this is still a **marginalization of a joint distribution**. It has the form

$$p(\mathbf{x}|\mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y}|\mathcal{T}).$$

The likelihood is **not** of the form $p(\mathbf{x}|\mathcal{T}, \mathbf{y})$

In the likelihood, $\mathbf{x}$ is the configuration of tip states we observe, and $\mathbf{y}$ is the configuration of internal node states we do not observe.

ANCESTRAL STATES AS LATENT VARIABLES

The basic form of this likelihood (a marginalization over latent/unobserved variables, $\mathbf{y}$),

$$p(\mathbf{x}|\mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y}|\mathcal{T}),$$

immediately suggests how we should determine ancestral states, $\mathbf{y}$.

Our best estimate for the ancestral state configuration is the particular $\mathbf{y}^*$ that makes the largest contribution to $p(\mathbf{x}|\mathcal{T})$.

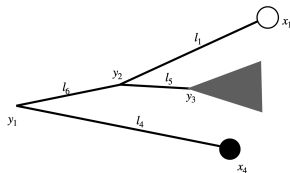
This is the **joint reconstruction** of ancestral states

This is a hard problem. For m traits and n tips in the tree, there are m^{n-1} different configurations of internal nodes. A generalization of the Viterbi algorithm makes it tractable³

³Tal Pupko et al. "A fast algorithm for joint reconstruction of ancestral amino acid sequences". In: *Molecular biology and evolution* 17.6 (2000), pp. 890–896.

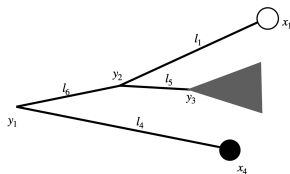
AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION

When we calculated the Likelihood, we moved from the tips up to the root, like so:



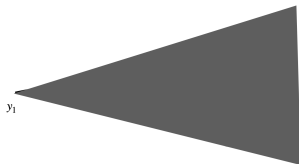
For this example tree, we would have conditional likelihoods $L_{y_3}(z)$, $L_{y_2}(z)$, and $L_{y_1}(z)$ for each value z (values of the possible ancestral states).

AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION



The key to finding the joint reconstruction is this: in addition to calculating these conditional likelihoods to combine them later, we store them separately for a second pass from the root back to the tips

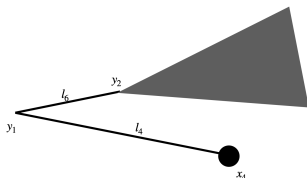
AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION



When we get to the root in the pruning algorithm, we will have conditional likelihoods $L_{y_1}(z_1)$, $L_{y_1}(z_2)$, ... $L_{y_1}(z_m)$ for each of the m possible states at the root.

Start by setting $\hat{y}_1 = \operatorname{argmax}_z \pi(z) L_{y_1}(z)$ as the most likely root state. We proceed back down to the tips:

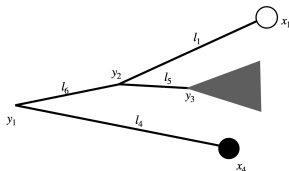
AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION



Next, we need to select the state at y_2 . To do so, we use the state we chose at the root, $\hat{y}_1$, and calculate $p_{\hat{y}_1, z}(l_6)L_{y_2}(z)$ for each z (a transition probability multiplied by the likelihood of the subtree descending from y_2).

We set $\hat{y}_2 = \operatorname{argmax}_z p_{\hat{y}_1, z}(l_6)L_{y_2}(z)$

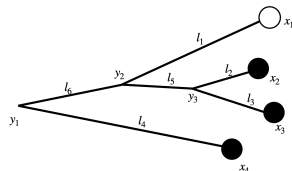
AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION



Next, we need to select the state at y_3 . We use the state we chose at the parent node, $\hat{y}_2$. We calculate $p_{\hat{y}_2, z}(l_5)L_{y_3}(z)$ for each z .

We set $\hat{y}_3 = \operatorname{argmax}_z p_{\hat{y}_2, z}(l_5)L_{y_3}(z)$

AN ALGORITHM TO FIND THE JOINT RECONSTRUCTION



Now, we have $\mathbf{y}^* = (\hat{y}_1, \hat{y}_2, \hat{y}_3)$ as our joint reconstruction. We had to calculate m quantities at each internal node, which means the calculation is linear in m (the number of states in the Markov chain).

If you have worked with Hidden Markov models before, you might recognize this algorithm as a generalization of the Viterbi algorithm.

MARGINAL RECONSTRUCTION OF ANCESTRAL STATES

So much for determining the most likely configuration of internal nodes, $\mathbf{y}^*$ (the **joint reconstruction**).

Alternatively, we may want to know the most likely state of an individual node, y_k , independent of the other nodes. This **marginal ancestral state reconstruction**⁴ proceeds by calculating, for each internal node y_k , the expression

$$p(\mathbf{x}, y_k | \mathcal{T}) = \sum_{y_1, y_2, \dots, y_{k-1}, y_{k+1}, \dots, y_n} p(\mathbf{x}, y_1, y_2, \dots, y_{k-1}, y_k, y_{k+1}, \dots, y_n | \mathcal{T}).$$

The value of y_k that corresponds to the largest $p(\mathbf{x}, y_k | \mathcal{T})$ is the **marginal ancestral character estimate** for node y_k .

⁴Dolph Schluter et al. "Likelihood of ancestor states in adaptive radiation". In: *Evolution* 51.6 (1997), pp. 1699–1711, Joseph Felsenstein. "Inferring phylogenies". In: *Inferring phylogenies*. 2004, pp. 664–664.

MARGINAL ANCESTRAL STATE ESTIMATION

$$L_{y_1} = \sum_{y_1} \pi(y_1) L_{y_1}(z).$$

The individual $L_{y_1}(z)$ terms in the likelihood suggest a way to carry out **marginal ancestral character estimation** for the root.

Just to reiterate: $L_{y_1}(z)$ is the likelihood of the tree conditional on the root taking on state z

The value of z that makes the largest contribution to L_{y_1} is the most likely state. We select it as our **marginal** estimate for the root state.

MARGINAL ANCESTRAL STATE ESTIMATION

So, we can find a marginal ancestral character estimate for the root of the tree, but what about all of the other nodes?

Whenever the Markov model of trait change is reversible (it usually is), the tree can actually be re-rooted at any node. So we just re-root, and then carry out the pruning algorithm each time⁵.

In this way, it is possible to calculate

$$p(\mathbf{x}, y_k | \mathcal{T}) = \sum_{y_1, y_2, \dots, y_{k-1}, y_{k+1}, \dots, y_n} p(\mathbf{x}, y_1, y_2, \dots, y_{k-1}, y_k, y_{k+1}, \dots, y_n | \mathcal{T}),$$

for each y_k . The value of y_k that gives the largest value for this expression is the marginal estimate.

⁵Joseph Felsenstein. "Inferring phylogenies". In: *Inferring phylogenies*. 2004, pp. 664–664.

JOINT VS. MARGINAL ANCESTRAL STATE ESTIMATION

Marginal character estimation is certainly tractable, but does not always produce the same estimates as the joint reconstruction⁶; they are designed to address different types of questions

In the ace function in R, there is an optional command “marginal” that defaults to FALSE, so that ace carries out a joint reconstruction by default

⁶Tal Pupko et al. “A fast algorithm for joint reconstruction of ancestral amino acid sequences”. In: *Molecular biology and evolution* 17.6 (2000), pp. 890–896.

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WORKING IN R

We will use the R package ape to simulate and fit these models to example datasets (and real ones) Make sure you have the following

R packages installed:

- ape
- phytools

WORKING IN R

In the practical, you will

- Fit Markov models to trees with tip states
- Simulate fitted Markov models
- Explore some example datasets